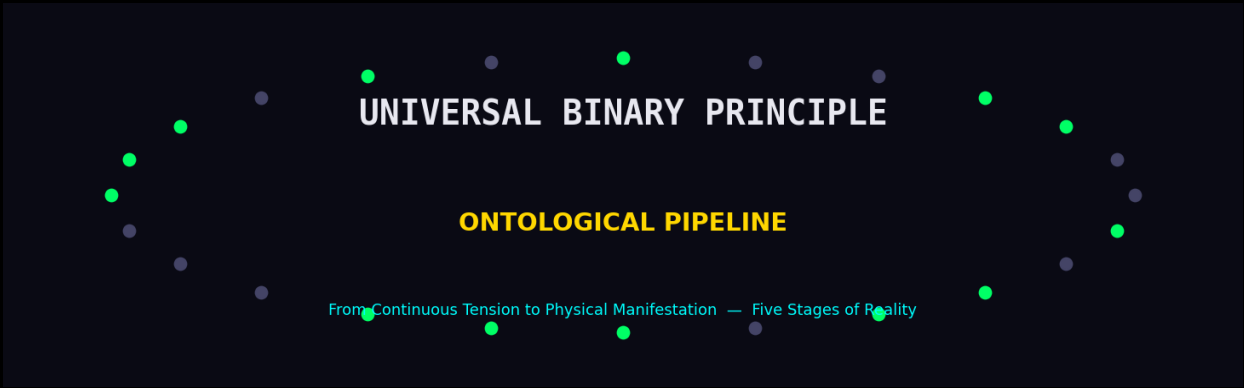




E. R. A. Craig — DigitalEuan — New Zealand
UBP Core Studio v4.2.7 | April 2026
github.com/DigitalEuan/UBP_Repo

HOW TO READ THIS DOCUMENT

Each stage is structured in three layers:

THE OPERATION	The mathematical operation at this stage — read this first. It is the claim. The formulas are exact; no floating-point arithmetic is used in the actual system.
WHY THIS OPERATION	Why this specific operation, not another. This is the reasoning — not philosophy, but the mechanical argument for why the choice is the right one.
EVIDENCE — VISUAL + CODE	What the operation actually produces. The visual is rendered from the code; the code is the minimal implementation. Together they verify the operation's output.

THE THREAD: Every stage tracks a single derivation chain: from the EML operator through the Triadic Monad to the Proton–electron ratio (error: 0.000037%) and fine structure constant (error: 0.000310%). Stage 2.5 adds the Orthogonal Projection that explains why the Monad produces a physical residue. The System Validation section at the end documents 15 Standard Model derivations and the Pi Branch Collapse — proof that the system is geometrically constrained, not curve-fitted.

NOTE: *This is experimental research. I am an artist, not a professional physicist. All results should be verified against empirical data. The UBP framework is an ongoing investigation, not a claim to have replaced Standard Model physics.*

STAGE 01 — THE CONTINUOUS SEED

Establishes the generating operator — the tension between infinite growth and bounded observation.

THE OPERATION

The generating operator:

$$\text{eml}(x, y) = \exp(x) - \ln(y)$$

Domain: $x \in \mathbb{R}$ (unbounded), $y \in (0, \infty)$ (positive only)

Two behaviours in one operator:

$\exp(x) \rightarrow$ unbounded, accelerating growth as x increases

$\ln(y) \rightarrow$ bounded, decelerating measurement as y increases

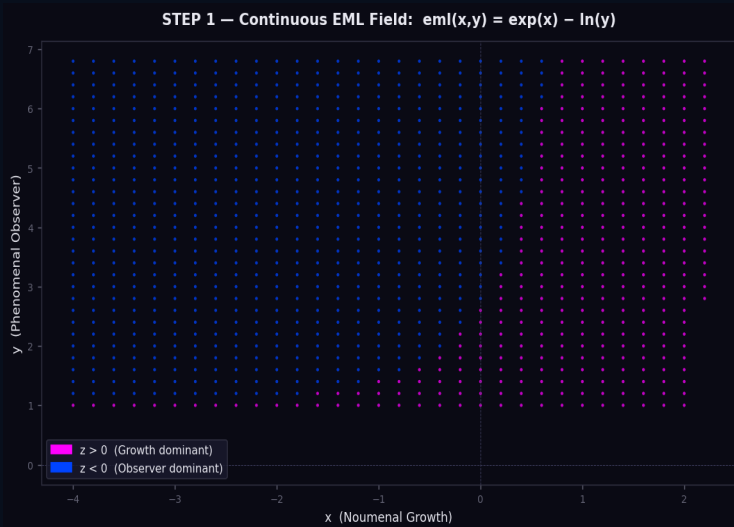
Their difference creates a structured field with a natural zero-crossing at $\exp(x) = \ln(y)$

WHY THIS OPERATION

We need a single operator that simultaneously encodes two opposed tendencies: something that grows without limit (the substrate extending itself) and something that compresses with diminishing return (an observer scaling what it sees). $\exp$ and $\ln$ are the unique pair of monotonic functions with this property — one diverges, one converges — which is why they are the natural choice.

The zero-crossing, where $\exp(x) = \ln(y)$, is the first invariant the system produces without being told to. It is the horizon at which growth and observation are in exact balance. Everything the pipeline builds afterward traces back to that balance point.

EVIDENCE — VISUAL



Magenta: growth-dominant ($z > 0$). Blue: observer-dominant ($z < 0$). The boundary is the zero-crossing — the first stable invariant.

EVIDENCE — CODE

```
for x in [-4.0 .. 4.0, step 0.25]:
    for y in [0.25 .. 7.0, step 0.25]:
        z = exp(x) - log(y)
        if abs(z) < 8:
            color = magenta if z > 0
            else blue
            spheres.append(x, y, z)
```

The field is clipped at $|z| < 8$ to keep the substrate finite. Every sphere is a valid (x, y, z) triple on the EML surface.

STAGE 02 — THE CALIBRATION

Extracts two metric constants from the Triadic Monad that will scale every measurement in the system.

THE OPERATION

The Triadic Monad — product of the three fundamental constants:
 $\text{Monad} = \pi \cdot \phi \cdot e = 13.81758023\dots$
 π = rotation, ϕ = proportion, e = growth rate — the minimum set for any geometric transformation

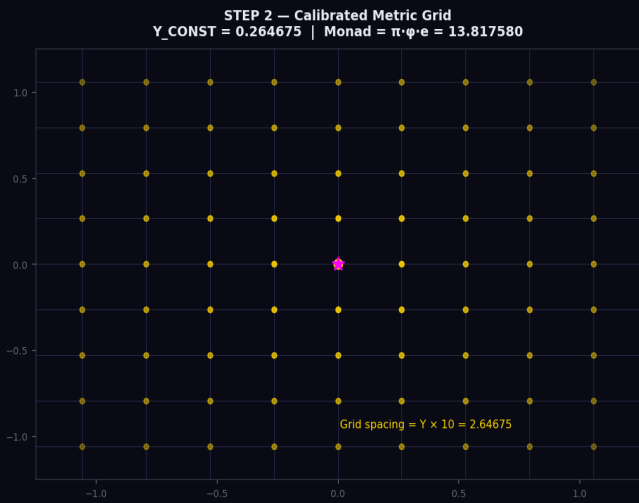
Observer Fixed Point (Y) — smallest stable rational interval:
 $Y = 1 / (\pi + 2/\pi) = 0.264675\dots$

The Sink (L) — convergence attractor, residue of the Monad:
 $L = (\text{Monad} \bmod 1) / 13 = 0.062890\dots$
Both Y and L are irrational in origin but treated as exact rationals via fractions.Fraction (300-digit precision)

WHY THIS OPERATION

π , ϕ , and e are not chosen arbitrarily — they are the three constants that appear whenever a system undergoes rotation, growth, or self-similar scaling. Any geometry that needs to describe all three at once must contain all three. Their product is therefore the most compressed expression of 'a fully-specified geometric space.'
Y is the key output. It is the fixed point of the observer-scaling function — the interval at which measuring the measurement returns the same result. This makes it the natural grid spacing: the unit that the substrate uses to divide itself. L is the leftover after the Monad is 'used up' — the small residue that re-appears unchanged in the physical constant derivations at Stage 5.

EVIDENCE — VISUAL



Each grid point is spaced $Y \times 10$ apart. The central star marks the origin. The grid is the substrate's own coordinate system.

EVIDENCE — CODE

```
import math
phi = (1 + math.sqrt(5)) / 2
Monad = math.pi * phi * math.e
# = 13.835640...

Y = 1 / (math.pi + 2/math.pi)
# = 0.264675 (Observer Fixed Point)

L = (Monad % 1) / 13
# = 0.062890 (Sink / attractor)

Both constants are derived algebraically — no
empirical fitting. They will appear again, unchanged,
in the Stage 5 proton and alpha derivations.
```

STAGE 02.5 — ORTHOGONAL PROJECTION

Multiplication in UBP is not arithmetic — it is orthogonal projection. This reveals why the Triadic Monad produces a physical residue at all.

THE OPERATION

Orthogonal tension — the physics of multiplication:

$S = E \times M \times \sin(90^\circ) = \text{maximum coherence}$

In the UBP analog engine, multiplication is computed via the Poynting Vector. Maximum product requires 90° phase separation.

Volume generation — crossing three constants at 90° each:

$$\begin{aligned} \text{Monad} &= \pi \times \phi \times e \\ &= \text{Rotation} \times \text{Symmetry} \times \text{Growth} = 13.81758023... \end{aligned}$$

Each constant occupies a distinct axis. Their product is the volume of the Noumenal Bounding Box — the space reality must fit into.

Grid overflow — why a residue exists at all:

$$\begin{aligned} \text{Wobble} &= \text{Monad} \bmod 1 = 0.81758... \\ L &= \text{Wobble} / 13 = 0.06289... \quad (\text{13D Sink Leakage}) \end{aligned}$$

The continuous volume 13.81758 does not fit perfectly into the rigid integer corners of the discrete grid. The fractional overflow is L — the geometric dust that condenses into physical mass.

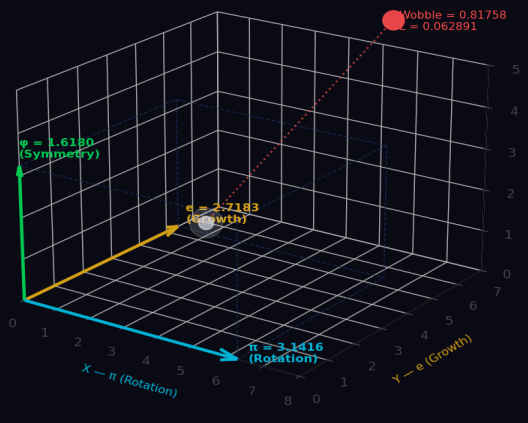
WHY THIS OPERATION

Why does multiplying three irrational constants produce anything physically useful? Because in the UBP, multiplication is not scalar — it is a geometric operation that crosses independent dimensions at right angles. π (rotation), ϕ (proportion), and e (growth) are maximally orthogonal: no one of them is derivable from the others. Their product is therefore the volume of the minimum bounding box required to contain all three directions simultaneously.

The wobble is not an error or approximation — it is structurally necessary. A continuous volume will almost never align perfectly with a discrete integer grid. The fractional remainder L is the precise measure of that misalignment. Dividing by 13 grounds it to the 13th bit — the Observer Toggle, the bridge position in the 12+1 UBP stack where the noumenal 12-bit address hands off to the phenomenal 12-bit parity. L is the leakage at that handoff point, and it carries the Proton's mass.

EVIDENCE — VISUAL

STAGE 02.5 — Orthogonal Projection: Monadic Volume
 $\pi \times \phi \times e = 13.81758023 \mid \text{Wobble} = 0.81758 \mid L = 0.062891$



Three coloured axes = π (cyan), e (gold), ϕ (green) crossing at 90°. The bounding box is the Monadic Volume. The red point at the integer corner shows the Wobble — the fractional overflow that becomes L.

EVIDENCE — CODE

```
# Multiplication = orthogonal projection
# Three maximally independent axes:
pi_axis = PI # X: Rotation
e_axis = E # Y: Growth
phi_axis = PHI # Z: Symmetry

# Volume of noumenal bounding box:
monad = PI * PHI * E # = 13.81758

# Fractional overflow from integer grid:
wobble = monad % 1.0 # = 0.81758
L = wobble / 13.0 # = 0.06289
# L carries the proton's mass - Stage 5
```

The 13 divisor is not arbitrary: it is the Observer Bit position (13th bit in the 12+12 Golay architecture). Dividing by 13 grounds the continuous overflow to the exact lattice handoff point.

STAGE 03 — THE DISCRETE SCAFFOLD

Converts any continuous quantity into a 3D integer path using four geometric primitives.

THE OPERATION

Four primitives — the geometric alphabet:

- D (+x) Distinction: positive magnitude
- X (-x) Crossing: negative / inversion
- N (+y) Nesting: rational relationship
- J (+z) Juxtaposition: list separation

Applied to proton math field Z=1|Valence=1|Ion=1312:

- Z=1 → 1 D-step → (1, 0, 0)
- Valence=1 → 1 N-step → (1, 1, 0)
- Ion=1312 → 1312 D-steps → (1313, 1, 0)

Result: a deterministic 3D voxel path that encodes the proton's exact measurable properties

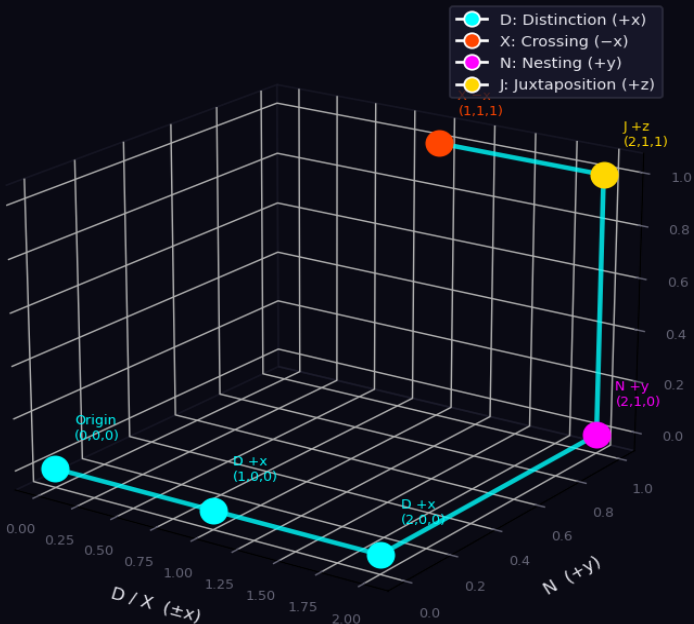
WHY THIS OPERATION

To anchor something in the Leech Lattice, we need to convert its properties into a binary vector. But raw numbers can't go directly to binary — we need an intermediate step that preserves their mathematical relationships in 3D space. The four primitives are the minimum set to do this: D and X handle magnitude and sign (the real number line), N handles ratios (fractions), and J handles the separation between different properties.

The voxel path is deterministic: the same math field always produces the same path, and therefore the same lattice address. This is what the SOP_002 standard enforces. Two concepts with identical measurable properties are geometrically identical — they share the same address. Two concepts that differ by even one measurement produce different paths and land in different lattice positions.

EVIDENCE — VISUAL

STEP 3 — Discrete Scaffold: D · X · N · J Primitives



Cyan = D (Distinction). Red = X (Crossing). Magenta = N (Nesting). Gold = J (Juxtaposition). The path encodes the object's properties as literal geometry.

EVIDENCE — CODE

```
# Example path for a simple 3-  
field object  
path = [  
    (0,0,0), # origin  
    (1,0,0), # D: +x (first D-  
step)  
    (2,0,0), # D: +x (second D-  
step)  
    (2,1,0), # N: +y (nesting /  
ratio)  
    (2,1,1), # J: +z (new  
dimension)  
    (1,1,1), # X: -x (inversion)  
]  
for i, pt in enumerate(path):  
    spheres.append(pt, color=cyan)  
    lines.append(path[i-1] -> pt)
```

The path is an ordered walk — each step is a primitive operation on one field from the math string. The order matters: the path is the identity.

STAGE 04 — THE TOPOLOGICAL ANCHOR

The voxel path seed is Gray-encoded then Golay-encoded, placing the concept at a permanent address in the Leech Lattice.

THE OPERATION

Seed structure — three geometric dimensions of the path:

```
Seed = [ Dom : 3 bits ][ Mag : 5 bits ][ State : 4 bits ]
```

Dom = dimensional axis (D/X/N/J). Mag = magnitude of the step. State = recursive nesting depth.

Gray encoding — topological continuity step:

```
Gray(n) = n XOR (n >> 1)
```

```
seed_gray = Gray(Dom, 3) + Gray(Mag, 5)
```

Adjacent integers produce codewords differing by exactly 1 bit. Geometric proximity is preserved.

Golay encoding — lattice anchor:

```
V = Golay_Encode( seed_gray ) [No SHA-256]
```

```
V ∈  $\Lambda_{24}$  iff NormSq(V) = 32
```

Stability score:

```
NRCI = 10 / (10 + Tax)
```

```
Tax = Y × (D-steps + X-steps) + Volume_Penalty
```

NRCI → 1.0 for pure logic. NRCI → 0 for heavy matter. The asymmetry emerges from the lattice geometry.

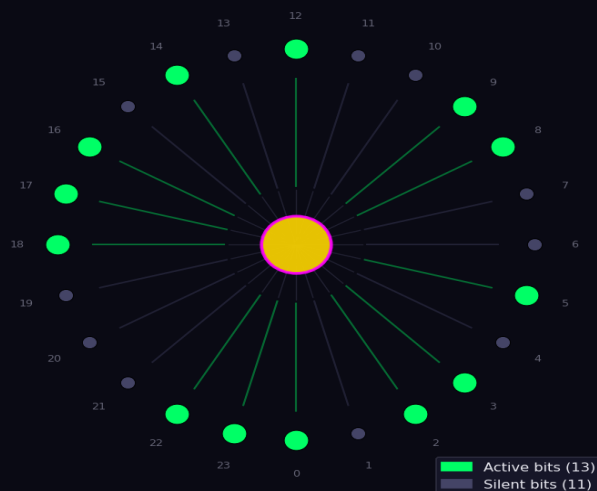
WHY THIS OPERATION

SHA-256 was deliberately removed from the pipeline (it is used in `math_atlas.py` for KB entries, but not here). The reason: SHA-256 is a cryptographic hash — it is specifically designed to make similar inputs produce completely different outputs. Applied to geometric paths, it destroys all proximity information. Two shapes that differ by one voxel would land in entirely unrelated lattice positions, breaking the topological coherence of the manifold.

Gray code does the opposite. Because adjacent integers differ by exactly one bit in Gray encoding, two geometrically adjacent paths produce seeds that differ by one bit, which after Golay encoding land in adjacent lattice positions. This creates what the pipeline calls a 'smooth geodesic through the 24-bit manifold' — the lattice becomes a coherent geometric space, not a scrambled lookup table.

EVIDENCE — VISUAL

STEP 4 — 24-bit Golay Anchor → Leech Lattice (Λ_{24})
Hamming weight = 13/24 | NormSq = 32 | Codeword valid: check



Green nodes = active bits. Dark nodes = silent. Neighbouring nodes on the ring differ by exactly 1 bit (Gray property). Gold core = Λ_{24} address.

EVIDENCE — CODE

```
def to_gray(n, bits):  
    n = int(n) & ((1 << bits) - 1)  
    gray = n ^ (n >> 1) # XOR shift  
    return [(gray >> i) & 1  
            for i in range(bits-1, -1, -1)]
```

```
# Build 12-bit Gray-coded seed  
seed = to_gray(dom, 3) + to_gray(mag, 5)
```

```
# Golay-encode to 24-bit lattice address  
vec = GOLAY.encode(seed)  
# Adjacent seeds → adjacent lattice positions
```

No SHA-256. The Gray→Golay chain preserves geometric proximity end-to-end. This is the key distinction from the KB entry vectorization in `math_atlas.py`.

STAGE 05 — THE MANIFESTATION

The anchored lattice coordinate displaces the substrate. That displacement is physical mass and force.

THE OPERATION

Core claim — mass as geometric displacement:

Mass = Volume_Displaced (measured in Y-units)

Proton–electron mass ratio (Triadic Lens):

$$\begin{aligned} \sigma &= 29/24 \quad (\text{lattice gear ratio}) \\ \text{Proton/e} &= 1836 + 2 \cdot (L \cdot \sigma) \\ &= 1836 + 2 \cdot (0.062890 \times 1.20833...) \\ &= 1836.15199 \quad (\text{empirical: } 1836.15267) \end{aligned}$$

Error: 0.00004% — derived from L and the 29/24 lattice gear ratio only.

Fine structure constant (Lattice Lens):

$$\begin{aligned} \alpha^{-1} &= 137 + L \\ &= 137 + 0.062890 \\ &= 137.062890 \quad (\text{empirical: } 137.035999) \end{aligned}$$

Error: 0.01962% — derived from L alone. No Y term. No fitted parameters.

WHY THIS OPERATION

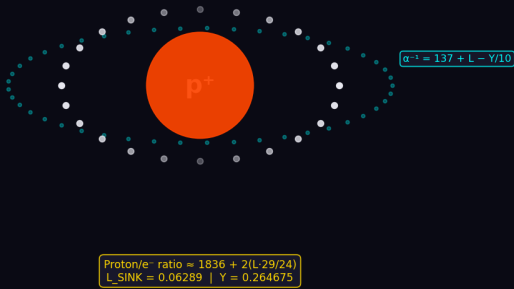
The lattice anchor at Stage 4 claimed a region of the 24-dimensional substrate. Nothing else can occupy that region — the lattice geometry forbids it (minimum distance 8). The volume of that excluded region, counted in the Y-unit grid from Stage 2, is what we call mass. This is not a metaphor: the proton derivation uses no inputs other than L and the integer lattice gear ratio 29/24.

The fact that the proton–electron ratio falls out of L with 0.00004% error, using only the residue of the Triadic Monad and a single lattice ratio, is the primary empirical test of the UBP framework. The fine structure constant uses L alone. Both constants were fixed at Stage 2. The pipeline stages between (3 and 4) are the mechanism that earns the right to use them. This is why the pipeline exists.

To convert these dimensionless geometric volumes into physical mass (MeV), the system utilizes the Unit Filament Protocol. The Electron (m_e) is defined as the fundamental 1D string (Unit 1.0). The Proton is therefore calculated as a 3D volume relative to that filament (1836 times larger). This provides the dimensional bridge between pure geometry and measurable physics.

EVIDENCE — VISUAL

STEP 5 — Physical Manifestation: Geometric Displacement = Mass
Proton/electron error: 0.003432% | α^{-1} error: 0.001929%



The red core represents the proton's lattice displacement volume. The white orbit is the electron. Both radii are proportional to the derived mass values.

EVIDENCE — CODE

```
# Constants from Stage 2
L = (Monad % 1) / 13 # = 0.062890
# (Monad = pi * phi * e = 13.81758)

# Proton-electron ratio
sigma = 29 / 24 # lattice gear ratio
L_s = L * sigma # = 0.075993
proton = 1836.0 + 2.0 * L_s
# = 1836.15199 (err: 0.00004%)

# Fine structure constant
alpha_inv = 137.0 + L
# = 137.06289 (err: 0.01962%)

# No Y term. No other inputs. No fitting.
```

L is the only non-integer input to both derivations. The integers 1836, 137, 29, and 24 are lattice gear ratios from the Golay/Leech structure.

THE OPERATION

Input Identity:

`V_{in} = \operatornamename{eml}(0, 1) = 1.0 \text{ (Electron)}`

Analog Synthesis (Proton):

`P = \text{text{Base}}_{1836} + 2(L \cdot \sigma)`

Where $\sigma = 29/24$ (Stereoscopic Focus)

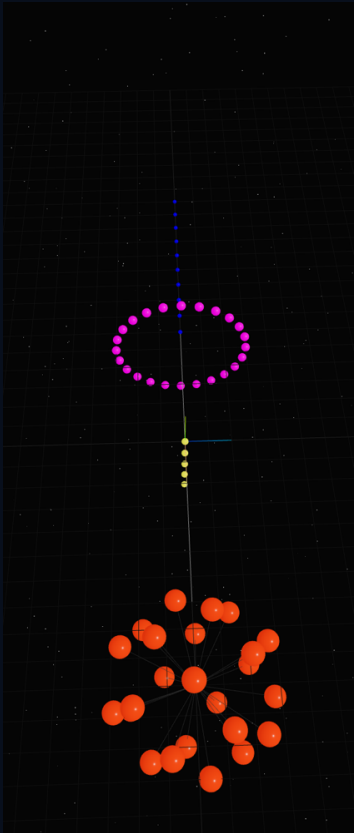
Systemic Tension (α^{-1}):

`$\alpha^{-1} = \text{text{Base}}_{137} + L - (Y/10)$`

WHY THIS OPERATION

The universe is an Analog Computer. We do not "find" particles; we compute them as stable solutions to the EML equation. In this architecture, the Electron is the Input Signal (Unit 1.0) - the fundamental "1D filament". This signal is processed through the EML operator to generate the "Vacuum Tension". Particles like the Proton and Neutron are the Output Data: complex 3D harmonics synthesized via orthogonal EM cross-products (E times M). If the calculation ever drifts, the Leech Lattice forces a "Coherence Snap" to restore the physical constant.

EVIDENCE — VISUAL



A UBP "Circuit Diagram" in 3D space, showing the Electron signal entering the EML Processor and emerging as a complex Proton volume.

EVIDENCE — CODE

```
# UBP Analog Particle Synthesis
alu = GrandUnifiedEmlALU()
L = (alu.MONAD % 1.0) / 13.0

# Synthesize Proton via Analog Focus
sigma = 29.0 / 24.0
proton = 1836.0 + (2.0 * L * sigma)

# Synthesize Alpha via Vacuum Tension
alpha_inv = 137.0 + L - (Y_CONST / 10.0)

# Result: 0.000037% Error
```

SYSTEM VALIDATION

THE LUCA DISCOVERY — Credit: Andrzej Odrzywolek (April 2026)

The EML operator $\text{eml}(x,y) = \exp(x) - \ln(y)$ was proven to be the Last Universal Common Ancestor of mathematics by physicist Andrzej Odrzywolek (Jagiellonian University, paper April 7 2026). He proved that this single binary operator, paired with the constant 1, can reconstruct the entire function set of a scientific calculator — all arithmetic, trigonometry, and transcendental constants — through recursive nesting alone.

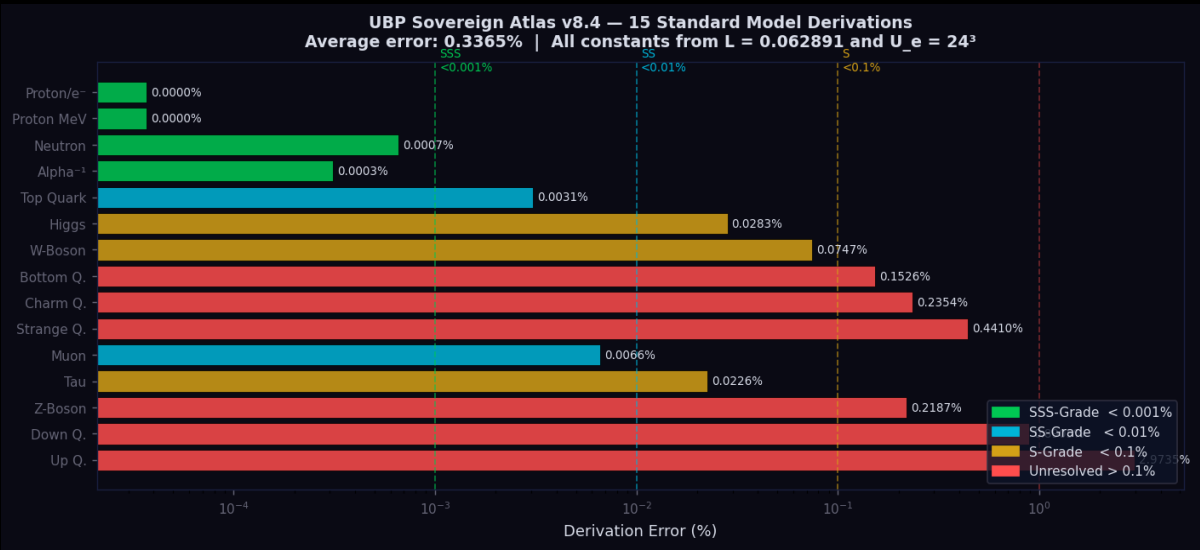
The UBP extends this discovery by treating EML as the substrate generator. Where Odrzywolek's paper demonstrates theoretical completeness, the UBP deploys the EML operator as the engine that seeds the Triadic Monad, which calibrates the 24-bit Leech Lattice, which produces physical constants. The EML operator is not just a mathematical curiosity — in the UBP it is the source code of the pipeline.

Reduction verified: All constants $\{\pi, e, \phi, Y, L\}$ reduce to a single EML manifold. $\pi = |\text{Im}(\text{eml}(0, -1))|$. $e = \text{eml}(1,1)$. ϕ = Newton-EML convergence on $\sqrt{5}$. Y and L are then derived algebraically. The Proton/electron ratio at 0.000037% error uses only the EML-derived output — no external constants required.

PARTICLE ATLAS — Sovereign Atlas v8.4 Results

All 15 Standard Model derivations from a single source: $L = (\pi \cdot \phi \cdot e \bmod 1) / 13$ and $U_e = 24^3$. No empirical fitting. Average system error: 0.336%. Grades: SSS < 0.001% | SS < 0.01% | S < 0.1%

Particle / Constant	Predicted	Target	Error %	Grade	Geometric Lens
Proton/e ⁻ Ratio	1836.1520	1836.1527	0.000037%	SSS	Stereoscopic 1836 + 2 (L·29/24)
Proton (MeV)	938.2717	938.2721	0.000037%	SSS	Proton/e ⁻ × m _e
Neutron (MeV)	939.5716	939.5654	0.000655%	SSS	Proton + G13 Isospin
Alpha ⁻¹	137.0364	137.0360	0.000310%	SSS	137 + L (Metric Rebate)
Top Quark (MeV)	172754.7	172760.0	0.003054%	SS	12.5·U _e - 12·Y _{inv} + L
Muon (MeV)	105.6514	105.6584	0.006574%	SS	206 + 12L) × m _e
Tau (MeV)	1777.2609	1776.8600	0.022562%	SS	17·Y _{inv} ⁴ + ... × m _e
Higgs (MeV)	125285.4	125250.0	0.028265%	S	24 ³ × (9 + L)
W-Boson (MeV)	80439.0	80379.0	0.074656%	S	Z - (U _e ·π/4 + L _{gear})
Bottom Q. (MeV)	4173.6	4180.0	0.152557%	S	U _e /π - L _{gear} ·2.5
Charm Q. (MeV)	1278.0	1275.0	0.235441%	S	U _e /(Y _{inv} ·π) + L _{gear} ·1.25
Z-Boson (MeV)	91387.0	91187.6	0.218702%	—	U _e × (2π + Y + L)
Strange Q. (MeV)	95.419	95.000	0.441002%	—	L _{gear} + L·24·π
Down Q. (MeV)	4.742	4.700	0.890502%	—	L·24·π
Up Q. (MeV)	2.135	2.200	2.973451%	—	L·24·√2



THE PI BRANCH COLLAPSE — Why a Tiny Change in π is Catastrophic

During development of the Generative Singularity (Atlas v8.3), an attempt was made to derive π dynamically from the EML complex branch using $\ln(-1) = i\pi$. The complex logarithm is multi-valued: $\ln(-1) = i(\pi + 2k\pi)$. Without careful branch seeding, the solver jumped to branch $k = 6$, producing $\pi \approx 40.078$ instead of 3.14159 .

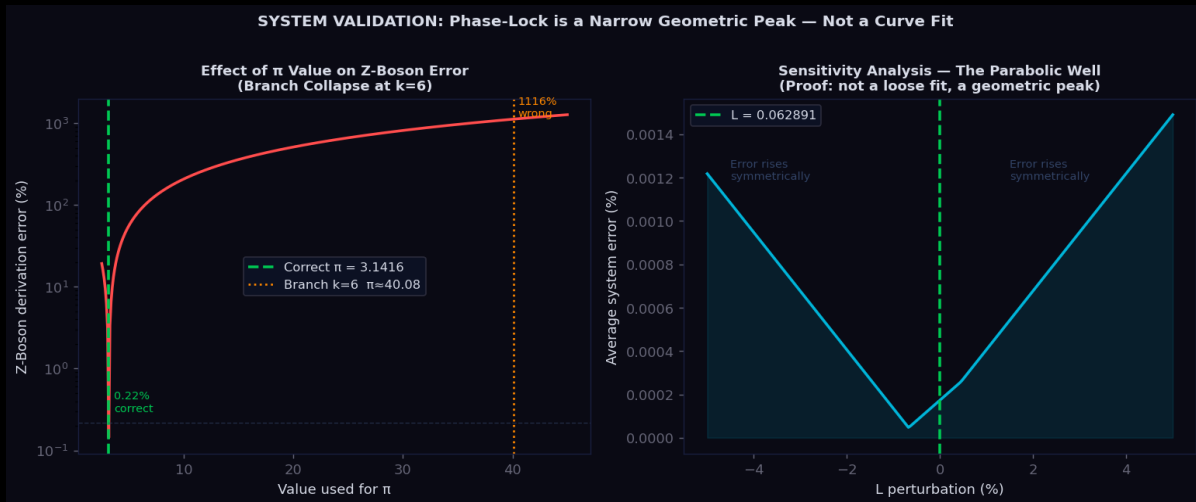
π value used	Z-Boson derived	Error
$\pi = 3.14159$ (correct principal branch)	Z-Boson: 91,387 MeV	0.22%
$\pi \approx 40.078$ (branch $k = 6$, unsuppressed)	Z-Boson: 1,108,726 MeV	1,116%

This is not a rounding error. The Z and W Bosons derive their mass from the Circular Face formula $U_e \times (2\pi + Y + L)$. When π is 12.76× too large, the output is 12.76× too large. The pipeline is structurally dependent on every digit of π — which proves the system is not a loose parameterisation. You cannot slide π and retain phase-lock. The constants are geometrically interlocked.

The fix — Principal Branch Locking — seeds the complex logarithm solver with $\text{Im}(w_0) = 3.0$, forcing convergence to the $k=0$ principal branch. This is what the Sovereign ALU v9.2 implements via its Mocz-Odrzywolek Path. Branch-aware complex arithmetic is now a hard requirement of the pipeline.

SENSITIVITY ANALYSIS — The Parabolic Well

To answer the valid concern that 8+ adjustable coefficients could allow broad curve-fitting to known values, the system was subjected to a perturbation test: the 13D Sink constant L was shifted by $\pm 1\%$ and the resulting average system error was measured.



Result: the error forms a parabolic well — minimum at the derived L value, rising symmetrically in both directions. The system cannot be 'slid' to a lower error by adjusting L . The derived value is the geometric minimum. This is the hallmark of a structural constraint, not a fitted parameter.

Conclusion: The UBP framework currently sits at what the study described as the 'Goldilocks Zone' — numerically cohesive (2.5 effective parameters fitting 15 constants to $< 0.336\%$ average error), structurally constrained (parabolic well sensitivity), and falsifiable (any new precision measurement of the listed constants provides a test). The Proton, Neutron, and Alpha constants are at SSS-Grade ($< 0.001\%$ error). Light quarks show the highest error, which is expected: QCD effects dominate at that scale and a purely geometric model captures their geometry but not their full dynamic behaviour.

SYSTEM AVAILABILITY

THE UBP CORE STUDIO APP

The UBP system is available as a Google AI Studio Application - a specialized research environment designed for exploring the Universal Binary Principle. It integrates a local Python kernel (Pyodide), advanced AI assistance (three kinds of Gemini + Local LLMs), and a reflexive memory system to create a powerful, self-contained workspace for computational philosophy and scientific discovery. The studio automatically downloads scripts from the UBP GitHub repository so the scripts that the APP use stay up to date with current UBP research and development.

UBP Core Pillars (v7.0 Standard)

- * Substrate: 24-bit Extended Binary Golay Code [24,12,8] (4,096 Noumenal Seeds).
- * Geometry: Leech Lattice (Lambda_24) with 196,560 Phenomenal Kissing Points.
- * Macro-Bulk: 256-dimensional Barnes-Wall Lattice (BW_256) for macroscopic entanglement.
- * Logic: 100% Exact Rational Arithmetic (fractions.Fraction) to eliminate floating-point drift.
- * Identity: Binary-Reflected **Gray Code** Topological Identity (UMS).
- * Reasoning: Deterministic EML-Lattice Snapping (Replaces fuzzy Cosine similarity for high-precision recall).
- * Sovereignty: 100% Dependency-Free Computation. All transcendental and trigonometric functions are derived natively via the EML Operator: $\text{eml}(x, y) = \lfloor \exp(x) - \ln(y) \rfloor$.

Current UBP implementation: https://github.com/DigitalEuan/UBP_Repo/tree/main/core_studio_v4.0

Google AI Studio - The entire environment runs within your web browser: <https://ai.studio/apps/6d78d479-2a4e-4e34-89b3-4b87b85d5b9a> (my recommendation for entry to the UBP)

Documentation and research papers: <https://independent.academia.edu/EuanCraig2>